

Sentinel

An Automated Electronic Climbing Belay Device

Project Proposal

Pol Mas Fillat

Norwegian University of Science and Technology (NTNU), Trondheim

Universitat Politècnica de Catalunya (UPC), Barcelona

Supervisor: Amund Skavhaug

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Abstract

Sentinel is a mechatronic system designed to function as an automated electronic climbing belay device – an “electric Grigri.” The device actively manages climbing rope slack during ascent and performs a fast-acting emergency brake in the event of a fall, removing the need for a human belayer. This document presents the initial project proposal for the Master’s Thesis, outlining the motivation, objectives, scope, and requirements of the system to be developed on an ESP32-based embedded platform.

1 Introduction

Rock and sport climbing rely fundamentally on the belay system: the mechanism and human partner responsible for managing rope slack and arresting a climber’s fall. Traditional belaying requires a trained, attentive human belayer holding the rope at all times, which introduces risks related to human error, fatigue, distraction, and inconsistent reaction times. Semi-automated mechanical devices, such as the Petzl GriGri, assist with braking but still require a human operator to feed rope and manage slack.

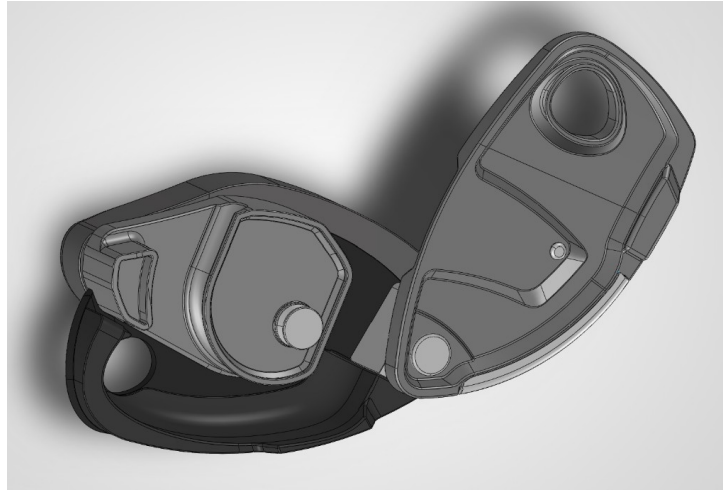


Figure 1: Reference CAD model of the Petzl GriGri mechanical assembly, used as a starting point for the design of Sentinel's rope-gripping mechanism.

The mechanical core of the Petzl GriGri, shown in Figure 1, is taken as the initial design reference for Sentinel. Its camming/pinching mechanism, which clamps the rope against a grooved wheel to generate braking friction, is a proven, compact, and mechanically reliable solution for assisted-braking belay devices. Rather than designing a rope-gripping mechanism from scratch, this project will use the GriGri's mechanical layout as a baseline and extend it with motor actuation, enabling the mechanism to be driven autonomously for both slack management (feeding and taking in rope) and emergency braking, rather than requiring a human to manually operate the cam lever and control the rope by hand.

Sentinel proposes a fully automated, sensor-driven belay device: an embedded mechatronic system capable of continuously monitoring rope position, tension, and dynamic motion in real time, and of autonomously regulating slack while remaining ready to trigger an emergency brake within milliseconds of detecting a fall event. By combining a rotary encoder, a load cell, and an inertial measurement unit (IMU) with a motor-actuated braking and rope-management mechanism, the system aims to provide a belay experience that is at least as safe and responsive as a competent human belayer, while removing the dependency on continuous human attention.

This project is developed as a Master's Thesis in Industrial Engineering at the Norwegian University of Science and Technology (NTNU), Trondheim, under the supervision of Amund Skavhaug, in collaboration with the Universitat Politècnica de Catalunya (UPC).

2 Main Objectives

1. Design and build a functional mechatronic prototype of an automated belay device capable of managing rope slack during a climber's ascent without human intervention.
2. Implement a real-time embedded control system, running on an ESP32 microcontroller, that fuses data from a rotary encoder, a load cell, and an IMU to estimate the climber's state (climbing, resting, falling).
3. Develop and validate a fast-acting emergency braking mechanism capable of arresting a fall within a safe reaction time and stopping distance, comparable to or better than manual belay

devices.

4. Create a remote control device / smartphone app / smartwatch app to allow the climber to control the belay device.
5. Characterize and document the mechanical, electrical, and control-software design through rigorous testing, in a form suitable for academic evaluation as a Master's Thesis.

3 Secondary Objectives

1. Explore adaptive slack management strategies (e.g. dynamic rope feed-out/take-in) to improve climber comfort and reduce unwanted rope drag or sudden tension spikes.
2. Investigate sensor fusion techniques (e.g. Kalman filtering) to robustly distinguish fall events from ordinary climbing dynamics (clipping, resting on the rope, lead falls).
3. Design the system with modularity and extensibility in mind, so that future work could explore additional features such as wireless connectivity, data logging for training analysis, or integration with gym safety infrastructure.
4. Evaluate the safety, reliability, and failure modes of the device against relevant climbing equipment standards (e.g. UIAA, EN 12841) as a reference framework, without necessarily seeking formal certification within the scope of this thesis.

4 Justification

Climbing gyms and outdoor climbing activities are experiencing sustained growth in popularity, and belaying remains one of the most safety-critical tasks in the sport. Human belayer error – including inattention, incorrect technique, and short-rope or hard-catch incidents – is a documented contributor to climbing injuries and fatalities. While auto-belay devices exist commercially (e.g. for top-rope-only gym use, using retractable cable/webbing systems), there is currently no widely available solution that automates belaying for lead climbing using a standard climbing rope while actively managing slack in real time.

An automated smart belay device addresses this gap by combining continuous, fatigue-free sensor monitoring with a mechanically robust braking system. From an engineering standpoint, the project is justified by the interdisciplinary challenge it poses: it requires the integration of mechanical design (rope-handling mechanism, braking system), embedded real-time software, sensor signal processing, and control theory, making it a comprehensive and appropriate capstone project for a Master's Thesis in Industrial Engineering.

5 Scope

The scope of this thesis project includes:

- Design and fabrication of a mechanical prototype capable of gripping, feeding, and braking a standard single climbing rope.

- Selection and integration of a rotary encoder, load cell, and IMU for real-time rope and motion sensing.
- Development of embedded C++ firmware on an ESP32 platform (using PlatformIO) implementing sensor acquisition, state estimation, slack-management control, and emergency-brake triggering logic.
- Bench and controlled-environment testing of the prototype, including simulated fall scenarios, to evaluate reaction time, braking distance, and false-trigger rate.
- Documentation of the design process, testing methodology, and results in the Master's Thesis document.

The following are explicitly **out of scope** for this thesis, unless revisited as future work:

- Formal certification of the device against climbing safety standards (e.g. UIAA, CE/EN).
- Outdoor field testing with real climbers at height.
- Mass-manufacturability and industrial design for a commercial product.

6 Requirements

6.1 Hardware Requirements

- **Microcontroller:** ESP32-based development board, providing sufficient processing power and I/O for real-time sensor fusion and motor control.
- **Rotary encoder:** to measure rope displacement and feed/retrieval speed.
- **Load cell (with amplifier, e.g. HX711 or similar):** to measure rope tension, used both for slack detection and as an input to fall detection.
- **Inertial Measurement Unit (IMU):** to detect dynamic motion characteristic of a fall event (e.g. sudden acceleration changes).
- **Actuator:** a servo or BLDC motor mechanism responsible for rope feed/take-in and for actuating the mechanical brake.
- **Braking mechanism:** a mechanically robust, fail-safe braking element capable of arresting fall loads.
- **Power supply:** suitable battery and power management circuitry for a portable prototype.
- **Structural frame/housing:** to mount all components securely around the rope path.

6.2 Software Requirements

- **Firmware development environment:** C++ using PlatformIO, targeting the ESP32 (`esp32dev`) platform.
- **Real-time sensor acquisition:** drivers/routines for reading the rotary encoder, load cell, and IMU at rates sufficient for fall detection latency requirements.
- **State estimation / sensor fusion:** algorithms (e.g. thresholding, filtering, or Kalman-based fusion) to classify the climber's current state and detect fall onset reliably.
- **Motor control logic:** closed-loop control of the rope-management actuator for slack take-up/feed-out.
- **Emergency brake control:** deterministic, low-latency logic to trigger the braking mechanism upon fall detection.
- **Documentation and version control:** structured project documentation (`docs/`) and development log (`docs/DEV_LOG.md`) tracking design decisions.

7 Expected Contributions

This thesis is expected to contribute:

1. A working proof-of-concept prototype demonstrating that automated, sensor-driven belaying with real-time slack management and fast emergency braking is technically feasible using accessible embedded hardware.
2. An open, documented reference architecture (hardware and firmware) for future research or development of automated belay devices, including sensor fusion strategies for fall detection.
3. Empirical performance data (reaction time, braking distance, false-positive/false-negative rates) characterizing the prototype's safety-relevant behavior under controlled test conditions.
4. A foundation for future work extending the system toward higher reliability, certification readiness, and real-world climbing gym deployment.